

Tirtho Mondal

📍 Kapilmuni, Khulna, Bangladesh | ✉ tirthomondal.2001@gmail.com | 📞 +8801571421684
🌐 LinkedIn | 🐙 GitHub | 🎓 Scholar | 🆔 ORCID | 🌐 Website

Career Objective

My goal is to contribute to higher education through effective teaching, student mentorship, and continuous learning in the field of Computer Science and Engineering. I aim to help students build strong academic and practical skills for successful careers while also contributing to research, innovation, and academic development.

Education

Khulna University of Engineering & Technology (KUET)	2022 – 2026
B.Sc. in Computer Science and Engineering	CGPA: 3.72 / 4.00
Kapilmuni College	2018 – 2020
Higher Secondary Certificate (H.S.C.) in Science	GPA: 5.00 / 5.00
Kapilmuni Sahachari Vidya Mandir	2016 – 2018
Secondary School Certificate (S.S.C.) in Science	GPA: 5.00 / 5.00

Research Interests

Optimization Theory | Game Theory | Computer Vision | Natural Language Processing | Machine Learning | Artificial Intelligence

Publications

1. D. Das, A. D. Das, F. Sadaf, A. Uddin, and **T. Mondal**, “Real-Time Assistive Navigation for the Visually Impaired: A Scalable Approach for Indoor and Outdoor Mobility,” *arXiv preprint arXiv:2504.20976*, 2025 [🔗](#)

Projects

3D Cafeteria Simulation 🔄	C++ / OpenGL 3.3 / GLFW / GLM
An interactive 3D cafeteria using texture mapping, Phong/Gouraud shading, Bézier curves, and animated characters.	
FPGA-Based Automatic Washing Machine 🔄	Verilog HDL / FPGA
7-state Moore FSM controller with timer, ALU rinse counter, and safety features on Basys 3 FPGA.	
Smart Navigation for the Visually Impaired 🔄	Python / Flutter
Assistive navigation system using computer vision and sensor fusion with real-time audio/haptic feedback.	
Adaptive UDP Congestion Control 🔄	C++ / OMNeT++
Genetic Algorithm based adaptive UDP packet-size control using real-time congestion feedback to reduce packet loss.	
Intelligent Bus Racing Simulation 🔄	Python / Pygame
5-lane racing simulation with AI buses using BFS/DFS pathfinding, Minimax, and fuzzy logic speed control.	
TutorFinder 🔄	Laravel / PHP / MySQL
Web platform connecting students with tutors through profile-based matching and scheduling.	
HouseHold Services 🔄	Android / Java
Android app for booking, tracking, and managing household service requests in real time.	
MobiCash 🔄	C++
Mobile banking simulation handling transactions, recharges, and user account management.	
Laundry Shop Management 🔄	SQL
Relational database for managing customers, orders, payments, and shop operations.	

Technical Skills

Languages	C, C++, Python, JavaScript, PHP, SQL, HTML/CSS, Java, Dart, Verilog HDL
Frameworks & Libraries	Laravel, Flutter, Android SDK, OpenGL, Pygame, JavaFX
Tools & Platforms	Git/GitHub, VS Code, IntelliJ IDEA, Android Studio, Jupyter, L ^A T _E X, OMNeT++, FPGA
Research & Data	NumPy, Pandas, OpenCV
Core Concepts	ML, Computer Vision, NLP, REST APIs, OOP, Computer Graphics, Digital Logic
Productivity	MS Word, Excel, PowerPoint

Leadership & Activities

Junior Executive, BitFest 2025, Dept. of CSE, KUET	2025
Class Representative, CSE Batch 20, Dept. of CSE, KUET	Sep 2024 – Feb 2025
Member, SGIPC, KUET	

Honours & Awards

- **Dean’s List:** Academic Years 2021–2022 and 2022–2023, Dept. of CSE, KUET
- KUET Technical Scholarship

Languages

Bengali: Native | **English:** Fluent